

AI ENGINEER CANDIDATE ASSIGNMENT

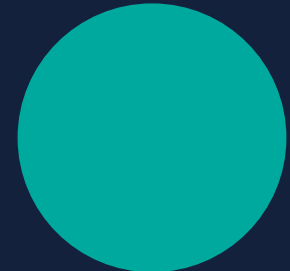
# AI-Powered Credit Risk Intelligence Platform

An end-to-end, explainable ML platform for loan default risk assessment, built on the Home Credit Default Risk dataset.

---

**Risha Zaidi**

MSc Data Science, CHRIST (Deemed to be University)





# Business Problem & Objective



## The Challenge

- Banks face pressure to make faster, more accurate, and explainable credit decisions.
- High-risk applicants often are not identified early enough in manual review workflows.
- Regulators and auditors increasingly expect risk decisions to be explainable, not opaque.
- Business teams often lack a fast way to explore applicant data without writing SQL.



## Platform Objective

- Predict the probability that a loan applicant will default.
- Convert probability into LOW / MEDIUM / HIGH risk bands.
- Compare multiple ML models and select the strongest performer.
- Let users explore applicant data through natural language (Talk-to-Data).
- Persist applicant data and prediction logs in SQLite.
- Deliver everything through an interactive Streamlit UI, deployed via Docker.



# Assignment Requirements & Scope



## EDA

Business insights & data quality analysis



## Machine Learning

Default probability prediction model



## Explainability

Understandable prediction reasoning



## Talk-to-Data

Conversational natural-language querying



## Business Rules

Business-readable decision insights



## User Interface

Multi-section Streamlit application



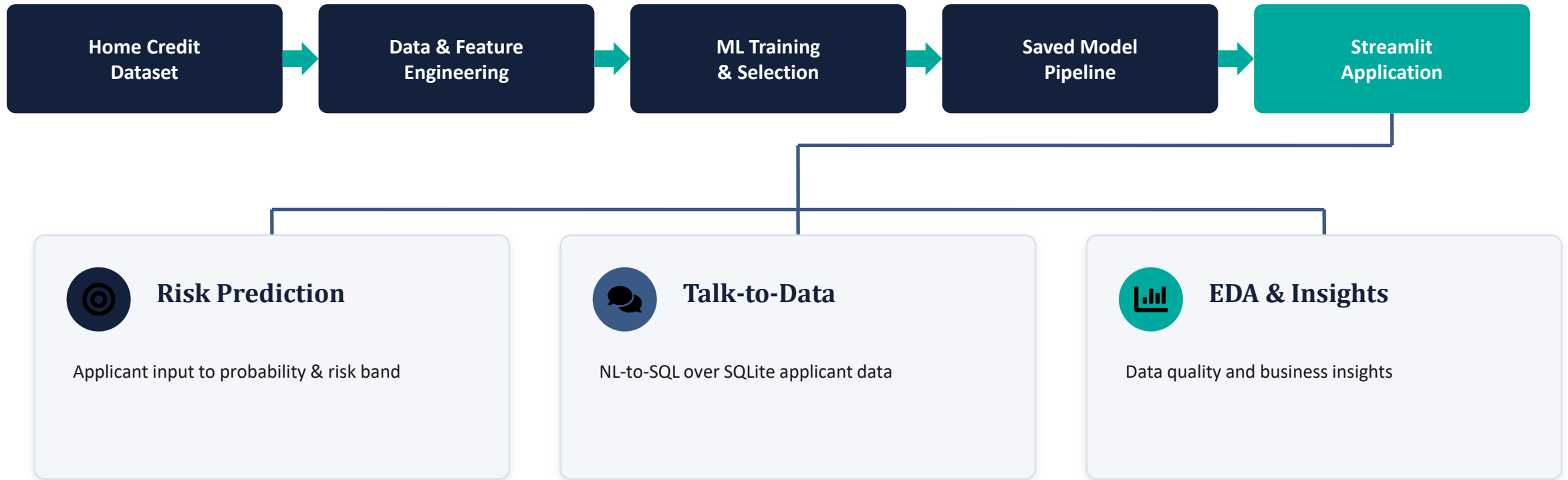
## Dockerization

Reproducible containerized deployment

*Evaluation weighting: ML Design 30% | LLM / Talk-to-Data 25% | EDA 15% | Prompt & Hallucination Control 15% | Dockerization 10% | Documentation 5%*



# Solution Overview



*One centralized feature-engineering layer and one risk-classification function are shared across training, prediction, and analysis - preventing drift between components.*



# Dataset & Data Understanding

## Home Credit Default Risk Dataset

300,000+

Original applicant records

122

Original raw columns

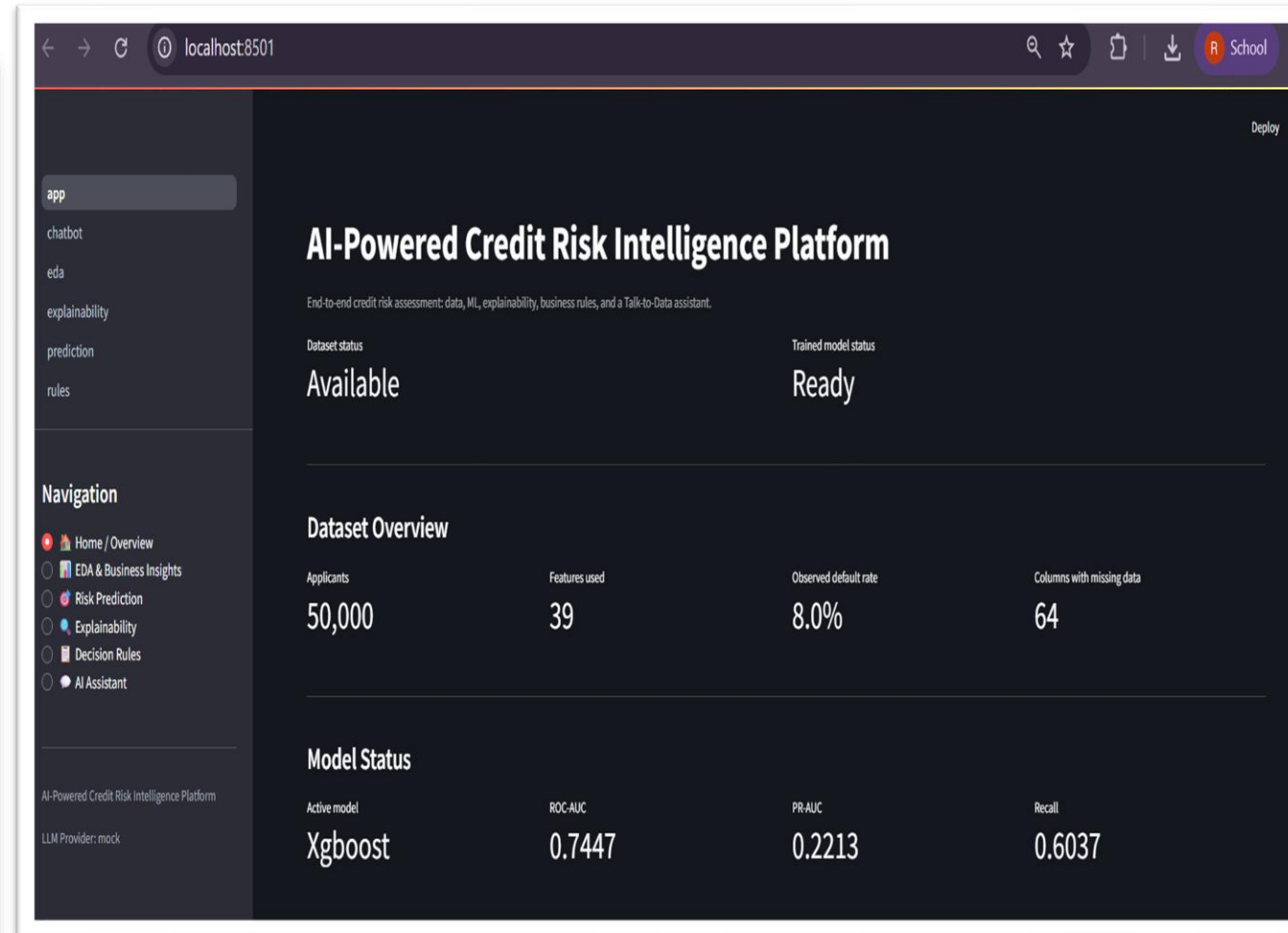
**TARGET**

0 = No Default | 1 = Default



### Engineering Note: Memory-Aware Sampling

Loading the full dataset on the local development machine hit memory limits. The pipeline supports a configurable DATA\_SAMPLE\_SIZE; a 50,000-row sample enabled successful local training and testing. Full-dataset training is supported in a higher-memory environment.





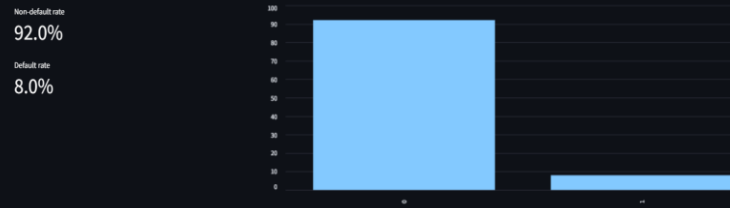
# EDA & Key Insights

At least 5 business insights are surfaced directly from the data. Insert the exact insight titles, figures, and supporting charts from your EDA run below.

1

## Target (Default) Distribution

Business Insight 1 — Target (Default) Distribution



Business interpretation: the dataset is heavily imbalanced — most applicants repay their loans. This is why the ML layer evaluates with ROC-AUC/PB-AUC and class-weighting rather than plain accuracy.

2

## External Credit Scores vs. Default

Business Insight 2 — External Credit Scores vs. Default



Business interpretation: applicants who defaulted (TARGET=1) tend to have lower average external credit scores than those who repaid — consistent with these scores being pre-existing risk indicators.

3

## Default Rate by Income Type

Business Insight 3 — Default Rate by Income Type



Business interpretation: default rates vary meaningfully across income types, from 0.0% to 100.0% — income type alone is not the whole story, but it's a useful segmentation for policy discussions.

4

## Credit-to-Income Ratio vs. Default

Business Insight 4 — Credit-to-Income Ratio vs. Default



Business interpretation: a higher credit-to-income ratio means the requested loan is large relative to the applicant's income — worth comparing between defaulters and non-defaulters as a leverage indicator.

5

## Missing Value Patterns

Business Insight 5 — Missing Value Patterns



Business interpretation: columns with high missingness (e.g. external scores or employment fields) often reflect applicants with thinner credit files — missingness itself can carry risk signal, not just noise.



# Data Processing & Feature Engineering

## Centralized Dataset Loader

- ✓ Dataset file existence & structural validation
- ✓ Required-column validation before use
- ✓ Memory-efficient, configurable-sample loading
- ✓ Configurable sampling via DATA\_SAMPLE\_SIZE
- ✓ Reproducibility enforced through RANDOM\_SEED

39

Modeling Features

27

Numerical

12

Categorical

*Feature engineering and cleaning logic is defined once and reused identically for training and inference.*



# Machine Learning Approach



## Logistic Regression

Interpretable linear baseline



## Random Forest

Bagged ensemble of decision trees



## XGBoost

Gradient-boosted ensemble



## Class Imbalance

In the 50,000-row training sample, only ~8.05% of applicants belong to the positive (default) class.

*A model predicting "no default" for everyone would score deceptively well on accuracy - accuracy is therefore not used for model selection.*



## Why PR-AUC

PR-AUC focuses on the precision-recall trade-off for the minority (default) class - exactly the class this platform must identify well.

**Selection order: PR-AUC then Recall (tie-break) then Inference speed (final tie-break).**





## Model Comparison & Results

| Model                     | ROC-AUC       | PR-AUC        | Precision     | Recall        | F1-Score      | Inference (ms/row) |
|---------------------------|---------------|---------------|---------------|---------------|---------------|--------------------|
| Logistic Regression       | 0.7479        | 0.2279        | 0.1605        | 0.6832        | 0.2600        | 0.101              |
| Random Forest             | 0.7446        | 0.2124        | 0.2015        | 0.5205        | 0.2906        | 0.9611             |
| <b>XGBoost (Selected)</b> | <b>0.7460</b> | <b>0.2318</b> | <b>0.1898</b> | <b>0.6075</b> | <b>0.2893</b> | <b>0.1889</b>      |



### XGBoost Selected

Ranked #1 on PR-AUC (0.2318) - the primary, imbalance-aware selection metric - ahead of Logistic Regression (0.2279) and Random Forest (0.2124).

*Selection strategy: rank by PR-AUC, tie-break on minority-class recall, then on inference speed. Accuracy is intentionally excluded given the class imbalance.*



# Credit Risk Prediction

Applicant Input

ML Pipeline

Default Probability

Risk Band

**LOW RISK**

Probability < 0.30

**MEDIUM RISK**

0.30 <= Probability < 0.60

**HIGH RISK**

Probability >= 0.60

## Credit Risk Prediction

Enter applicant details to get a default-risk assessment.

### Demographics

Gender

F

Number of children

0

Family members

2

Family status

Married

Age (years)

35

### Financial

Annual income

90000

Requested credit amount

600000

Loan annuity

27000

Goods price

540000

Owns a car

N

Owns real estate

N

### Employment & Credit Signal

Income type

Working

Occupation

Laborers

Years employed (0 if not employed)

5

External credit score #1

0.50

0.00

1.00

External credit score #2

0.50

0.00

1.00

External credit score #3

0.50

0.00

1.00

Assess Risk

## Result

**MEDIUM RISK** — 49.4% default probability

See the Explainability and Rules pages for why this applicant received this classification.



# Talk-to-Data System



## Mock Provider (Default)

Deterministic, offline, zero API cost. Covers all required query patterns for development, testing, and demonstration.



## Anthropic Provider (Optional)

Enabled via LLM\_PROVIDER, ANTHROPIC\_API\_KEY, ANTHROPIC\_MODEL - no code changes required to switch.

The screenshot displays the user interface of the Talk-to-Data System. It shows a series of queries and their corresponding results, each with a 'Generated SQL' button. The queries are:

- Query 1:** The minimum number of children recorded for an applicant is 0.
- Query 2:** What is the average number of children?
- Query 3:** The average number of children per applicant is 0.42.
- Query 4:** How many occupation groups are there?
- Query 5:** There are 18 distinct occupation groups in the dataset.
- Query 6:** Which occupation groups have the highest default rate?

The final query result is titled "Occupation groups by observed default rate" and lists the top 5 groups:

1. Low-skill Laborers — 15.57%
2. Waiters/barmen staff — 11.84%
3. IT staff — 11.54%
4. Drivers — 11.15%
5. Security staff — 10.69%

The highest observed default rate in this ranking is for Low-skill Laborers, at 15.57%.



# Prompt Optimization & Hallucination Control



## Schema-Aware Querying

The model only sees the actual applicant table schema - not the full database.



## SQL Validation Before Execution

Every generated query is parsed and validated before it ever touches SQLite.



## Read-Only Execution

Only SELECT statements are permitted; write/DDL operations are rejected outright.



## Restricted, Business-Relevant Schema

Only a curated, business-relevant column subset is exposed to the chatbot.



## Row Limits on Results

Returned rows are capped before being passed further down the pipeline.



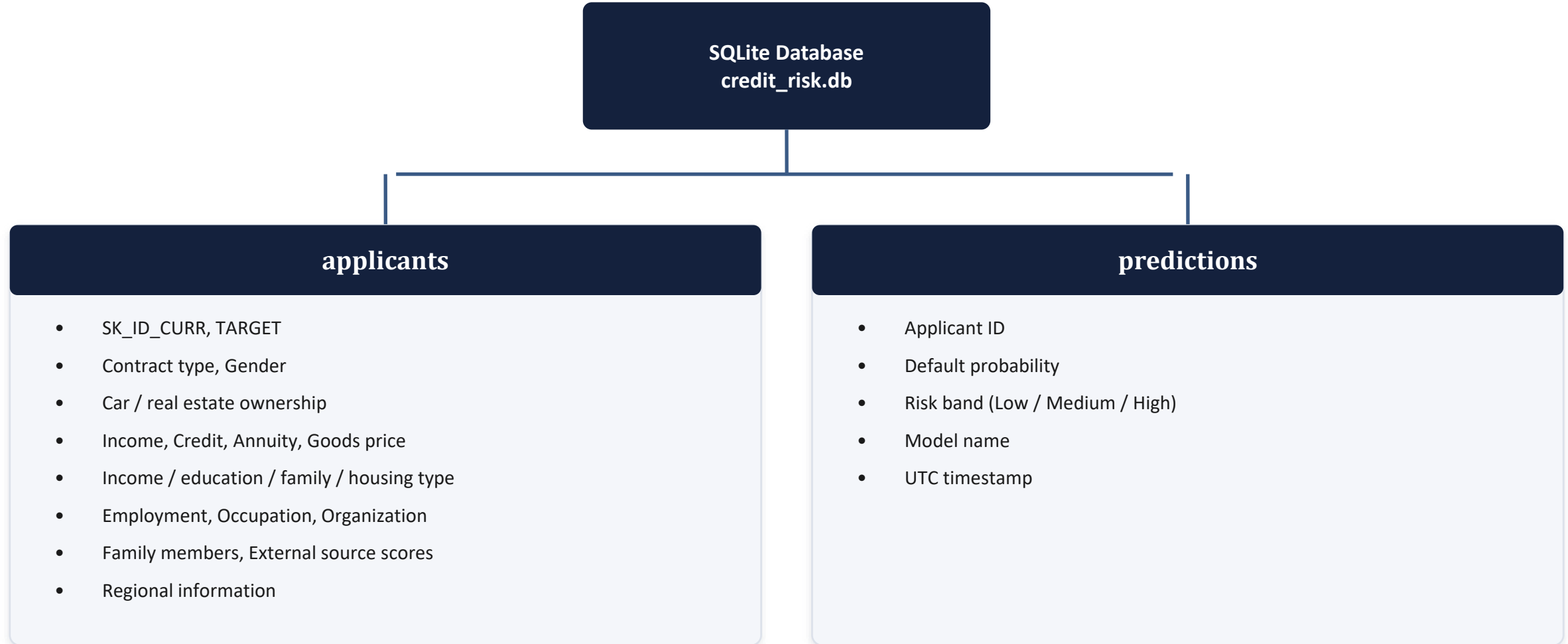
## Conversation Memory (3 Turns)

CONVERSATION\_MEMORY\_TURNS=3 bounds context - no unbounded history growth.

*Deterministic mock provider keeps supported query patterns fully predictable - no hallucinated SQL is possible for patterns it explicitly implements.*



# Database & System Design





# Explainability & Business Understanding

## Current Explainability Approach

- SHAP (TreeExplainer / LinearExplainer) computes global feature importance - mean absolute SHAP value - across a sample of applicants.
- Local, per-applicant SHAP values identify the top factors increasing and reducing that specific prediction.
- Technical feature names are translated into business-readable labels (e.g. EXT\_SOURCE\_2 -> "External credit score #2").
- If SHAP computation fails for any reason, the interface shows a clean fallback rather than a fabricated explanation.

### Prediction Explainability

Understand which factors influence the model globally and why it produced the latest applicant's risk prediction.

**MEDIUM RISK** — 49.4% default probability

Probability of Default  
**49.4%**

Model confidence (distance from the 50% decision threshold): 1%. This measures how far the prediction is from the point of maximum uncertainty and is not a calibrated statistical confidence interval.

The model estimates a 49.4% probability of default, placing this applicant in the MEDIUM RISK category.

### Local Explanation – Last Prediction

This applicant was classified as MEDIUM RISK with a 49.4% predicted probability of default.

#### Risk Contribution Breakdown

Each bar shows how strongly an individual feature pushes this specific prediction toward lower or higher predicted default risk.





# Dockerized Deployment

## Configuration

- Python 3.11-slim base image
- Dependencies installed, project copied, port 8501 exposed
- Built-in Streamlit health check
- Compose: port 8501:8501, .env loaded
- Volumes: ./data and ./models mounted from host
- restart: unless-stopped

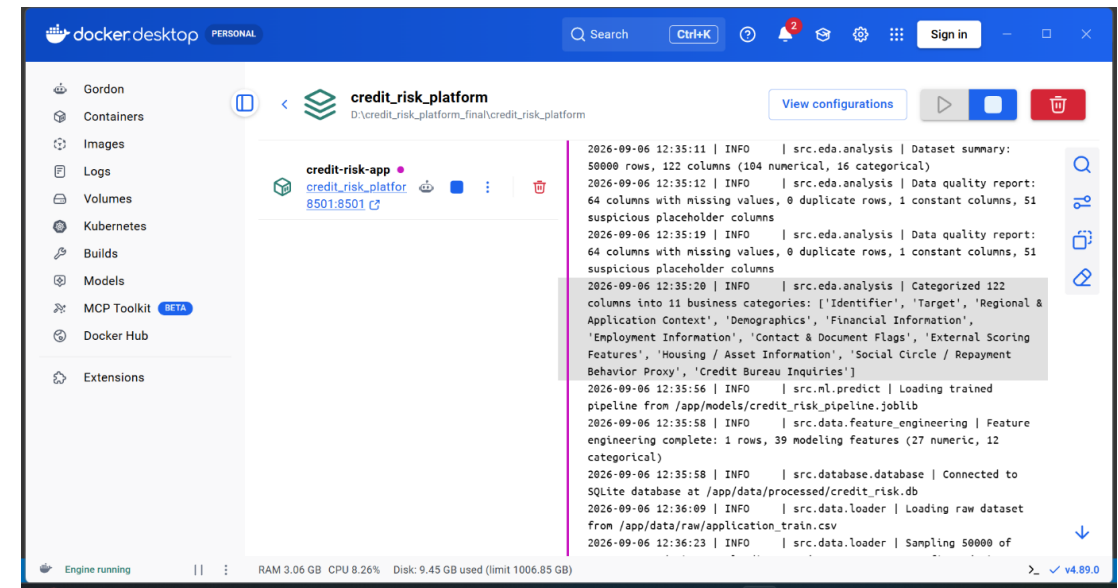
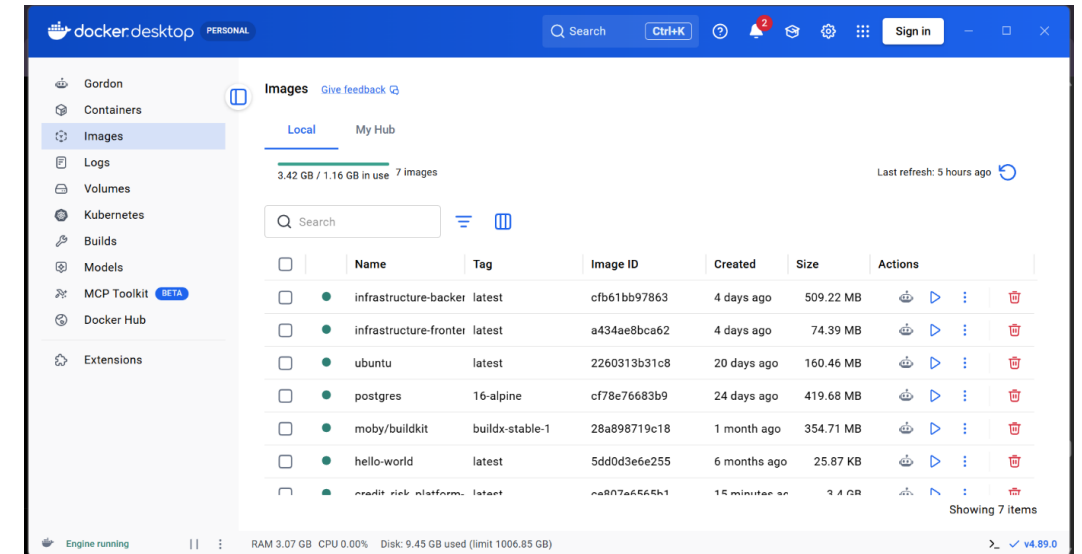
**docker compose up --build**



**Container Starts**



**App Available on localhost:8501**





# Engineering Challenges & Solutions

| Challenge                                               | Solution                                                                |
|---------------------------------------------------------|-------------------------------------------------------------------------|
| Memory limits loading the full 300,000+ row dataset     | Configurable DATA_SAMPLE_SIZE; successful 50,000-row local training run |
| Heavy class imbalance (~8% positive class)              | class_weight / scale_pos_weight plus PR-AUC-driven model selection      |
| Making applicant data accessible to non-technical users | Talk-to-Data natural-language querying interface                        |
| Safe natural-language-to-SQL querying                   | Restricted schema, SQL validation, and read-only execution              |
| Reproducible, portable deployment                       | Dockerfile + docker-compose with host-mounted data and models           |





# Testing, Architecture & Project Quality

35 / 35

**Automated Tests Passed**

*1 non-blocking warning observed during the run*

## Engineering Quality Highlights

- Modular source layout: src/data, ml, database, talk\_to\_data, utils
- Saved, versioned model artifacts (pipeline + metadata + comparison)
- SQLite persistence for applicant and prediction data
- Centralized, environment-based configuration (.env)
- Fully Dockerized deployment with health checks



# Final Outcome, Limitations & Future Enhancements



## Delivered

- EDA with business insights
- ML model comparison & selection
- SHAP explainability
- Talk-to-Data (NL-to-SQL)
- Streamlit multi-section UI
- Dockerized deployment



## Known Limitations

- Trained on 50,000-row local sample
- Default provider is deterministic mock
- Risk thresholds not cost-calibrated
- No production security hardening
- No fairness/bias audit yet



## Future Enhancements

- Full-dataset training (higher memory)
- Hyperparameter optimization
- Additional explainability (LIME)
- Business-cost threshold calibration
- Real-LLM-only mode, cloud deploy
- CI/CD and monitoring

# Thank You

Questions?

---

**AI-Powered Credit Risk Intelligence Platform**

Risha Zaidi | MSc Data Science, CHRIST (Deemed to be University)